

VƯƠNG VIỆT HÙNG

PERSONAL DETAILS

Full name:	Vương Việt Hùng	Languages:	Vietnamese (Native), English (IELTS 7.5)
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EDUCATIONAL QUALIFICATIONS

- **Soongsil University** Seoul, South Korea
Master's Degree in Electronic Engineering March 2024 – March 2026
 - GPA: 4.24/4.50
 - Thesis: Scene Flow-Guided Online Continual LiDAR Panoptic Tracking
 - Thesis length: 10,387 words
- **Hanoi University of Science and Technology** Hanoi, Vietnam
Bachelor's Degree in Control Engineering and Automation August 2019 – September 2023
 - GPA: 3.34/4.00
 - Thesis: Stereo Matching and 3D Reconstruction
 - Thesis length:

EMPLOYMENT HISTORY

- **VinDynamics** Hanoi, Vietnam
SLAM & Navigation Engineer March 2026 – current
Developed and deployed SLAM and height mapping systems for humanoid robotic platforms, with a focus on sensor integration, system evaluation, and optimization.
- **ANDA Lab, Soongsil University** Seoul, South Korea
Research Assistant March 2024 – March 2026
Conducted research in LiDAR panoptic segmentation, scene completion, domain adaptation, and continual learning; contributed to peer-reviewed publications, researcher mentoring, and technical seminars.
- **Rikai Mind** Hanoi, Vietnam
AI Engineer Intern August 2022 – January 2023
Developed and deployed a computer vision system for object volume estimation from monocular images and camera-object distance.
- **Mandevices Laboratory, Hanoi University of Science and Technology** Hanoi, Vietnam
Research Assistant August 2020 – September 2023
Conducted research in deep-learning-based bearing fault diagnosis and contributed to data collection, publications, and researcher mentoring.

RESEARCH OUTPUTS

J=JOURNAL, C=CONFERENCE, S=IN SUBMISSION

Published Research:

- [J.1] **Vuong, H. V.**, & Yoo, M. (2026). Domain adaptive LiDAR panoptic segmentation via pseudo-supervised instance learning. *IEEE Transactions on Intelligent Transportation Systems*. <https://doi.org/10.1109/TITS.2026.3695516>

Journal Impact Factor: 9.1

- Proposed PILLA, a domain adaptation framework that generates instance-level pseudo-labels via unsupervised clustering and applies a noise-aware loss for robust pseudo-supervision.
- Achieved 35.97% PQ on nuScenes in cross-dataset adaptation from SemanticKITTI, outperforming AdaptLPS by 13.14 percentage points.

[J.2] Ha-Phan, N.-Q., **Vuong, H. V.**, & Yoo, M. (2025). Instance embedding as queries for DETR-based LiDAR panoptic segmentation. *IEEE Transactions on Intelligent Vehicles*, 10(10), 4629–4641.

<https://doi.org/10.1109/TIV.2024.3488035>

Ha-Phan, N.-Q. and Vuong, H. V. contributed equally to this work.

Journal Impact Factor: 14.3

- Proposed a DETR-based architecture that uses instance embeddings as object queries for LiDAR panoptic segmentation.
- Achieved 63.40% PQ on SemanticKITTI and 77.08% PQ on nuScenes, outperforming previous methods by 1.20 and 1.94 percentage points, respectively.

[J.3] Nguyen, K. N. M., **Vuong, H. V.**, Ha-Phan, N.-Q., & Yoo, M. (2025). FuPaSCo: Long-range and local context fusion for 3D panoptic scene completion. *Image and Vision Computing*, 163, 105776.

<https://doi.org/10.1016/j.imavis.2025.105776>

Journal Impact Factor: 4.2

- Developed FuPaSCo, a dual-branch CNN–Transformer framework that combines local and long-range features through feature fusion and alignment for LiDAR panoptic scene completion.
- Achieved 26.10% $PQ^\dagger$ on SemanticKITTI, outperforming PaSCo by 2.49 percentage points.

[J.4] Nguyen, T. H., **Vuong, V. H.**, Dao, D. T., Tran, T. T., & Hoang, S. H. (2024). Generalized simulation-based domain adaptation approach for intelligent bearing fault diagnosis. *Arabian Journal for Science and Engineering*, 49(12), 16941–16957. <https://doi.org/10.1007/s13369-024-09282-1>

Journal Impact Factor: 3.1

- Developed a simulation-based domain adaptation approach that generates synthetic vibration signals from bearing geometry to address data scarcity in intelligent bearing fault diagnosis.
- Achieved 74–98% fault classification accuracy across different bearing types using a CNN trained on synthesized vibration signals.

[C.1] Nguyen, T. H., Le, H. A., **Vuong, V. H.**, Tran, L. D. A., Dao, D. T., Tran, T. T., & Hoang, S. H. (2024). ConvMamba: A data-efficient neural network for bearing fault diagnosis. In *2024 7th International Conference on Green Technology and Sustainable Development (GTSD)* (pp. 165–172). IEEE. <https://doi.org/10.1109/GTSD62346.2024.10674880>

- Developed ConvMamba, a data-efficient architecture combining convolutional networks and Vision Mamba for bearing fault diagnosis.
- Improved classification accuracy by 27 percentage points over the predecessor Hybrid-CNN-MLP model on the CWRU bearing fault dataset.

Manuscripts Under Review:

[S.1] Pham, T. P., **Vuong, H. V.**, Ha-Phan, N.-Q., & Yoo, M. (2026). SM3-PSC: Sparse multi-modal Mamba for 3D panoptic scene completion with multi-scale feature refinement. *Manuscript submitted for publication to IEEE Transactions on Image Processing*.

Journal Impact Factor: 15.3

- Developed SM3-PSC, a sparse multi-modal framework that fuses RGB semantics with LiDAR voxels through deformable cross-attention and uses Mamba for efficient long-range context modeling.
- Achieved 28.20% $PQ^\dagger$ on SemanticKITTI, outperforming FuPaSCo by 2.10 percentage points while running the 3D pipeline $2.8\times$ faster.

[S.2] Nguyen, K. N. M., **Vuong, H. V.**, Ha-Phan, N.-Q., & Yoo, M. (2026). LW-PSC: Lightweight design for LiDAR panoptic scene completion. *Manuscript submitted for publication to Applied Soft Computing*.

Journal Impact Factor: 7.8

- Developed LW-PSC, a lightweight BEV-based framework that combines Cartesian and Polar representations with center-focused feature refinement for real-time LiDAR panoptic scene completion.
- Achieved 13.59% PQ at 11.3 FPS on SemanticKITTI, outperforming PaSCo by 1.47 percentage points while reducing the model size from 40.65M to 29.16M parameters.

PROJECTS

- **Visual SLAM Profiling and Latency Optimization** August 2026
VinDynamics
 - Profiled the OKVIS2 visual SLAM pipeline, identifying landmark matching and local-window optimization as major bottlenecks, and evaluated low-latency front-end pose publishing with only 1.16 cm translational APE RMSE relative to back-end poses.
 - Redesigned frame scheduling to prioritize the latest available frame and drop stale frames, reducing average front-end latency from 124.00 ms to 47.00 ms and back-end latency from 164.30 ms to 87.64 ms.
- **Multi-Sensor Local Height Mapping for Humanoid Robot** July 2026 – August 2026
VinDynamics
 - Extended a LiDAR-based GPU height mapper to integrate Livox LiDAR and Intel RealSense depth data, implementing sensor synchronization, rate limiting, pose lookup, and robot self-filtering for deployment on an in-house humanoid robot.
 - Implemented and benchmarked pose extrapolation to compensate for SLAM latency, reducing translational APE RMSE from 2.148 cm to 0.445 cm (79.3%) for a 30 Hz depth camera using linear extrapolation.
- **Scene Flow-Guided Online Continual LiDAR Panoptic Tracking** September 2025 – February 2026
Graduation thesis, Soongsil University
 - Proposed SF-LPT, the first online continual learning framework for LiDAR panoptic tracking, using self-supervised scene flow for continuous adaptation and temporal instance association across sequential scans.
 - Achieved 16.45% LSTQ on SemanticKITTI in cross-dataset continual adaptation from nuScenes, outperforming CoDEPS by 5.87 percentage points.
- **Stereo Matching and 3D Reconstruction** March 2023 – August 2023
Graduation thesis, Hanoi University of Science and Technology
 - Implemented and compared block matching and deep learning approaches for dense stereo matching on SceneFlow (EPE: 77.32 px, D1: 79.04%).
 - Reconstructed 3D scenes from estimated disparity maps via triangulation and deployed the pipeline on a Jetson Nano-based wheeled robot.
- **Object Volume Estimation from Images** August 2022 – December 2022
Rikai Mind
 - Developed an object volume estimation model from monocular images and camera–object distance, achieving an R^2 score of 84.78%.
 - Extracted visual embeddings with CNNs, localized objects using keypoint regression, and fused visual features with camera–object distance for volume regression; deployed the model using FastAPI.

SKILLS

- **Programming languages:** Python, C/C++, MATLAB
- **Frameworks:** PyTorch, TensorFlow, OpenCV, ROS2, Kalibr (for sensor calibration)
- **Hardware platforms:** Jetson Orin Nano, Jetson Nano, Raspberry Pi
- **Sensors:** Intel RealSense camera, ZED X camera, Livox LiDAR
- **Development and deployment tools:** Git, GitHub, GitLab, Docker, FastAPI, Milvus
- **Others:** LaTeX